Experiment 1

Load Flow Analysis Using PSS®E

Lab Report

Student Name HAMZA AHMAD
Registration Number 2022-EE-165
Section B
Power System Operation and Control Laboratory
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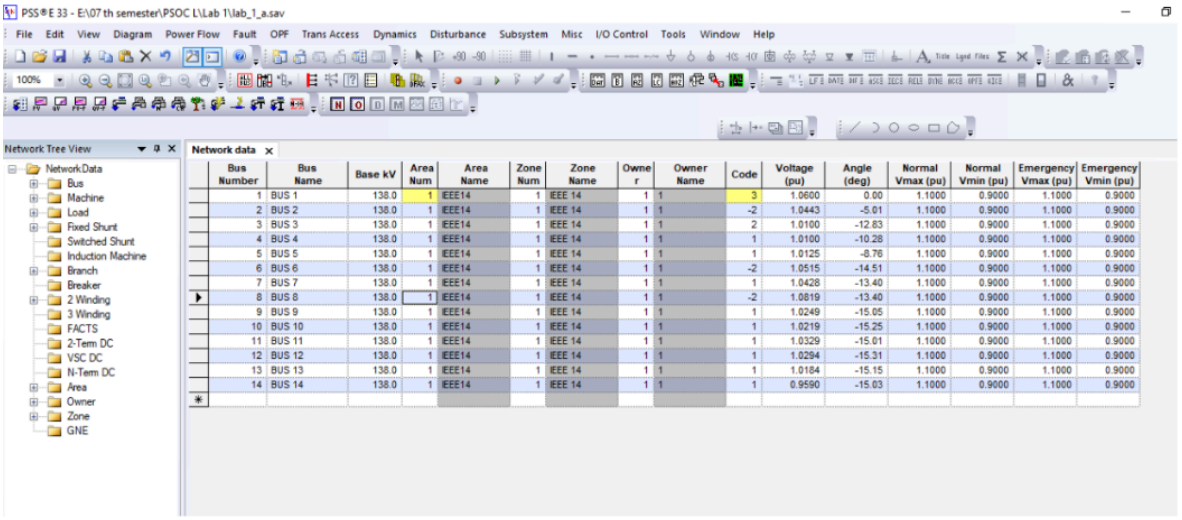
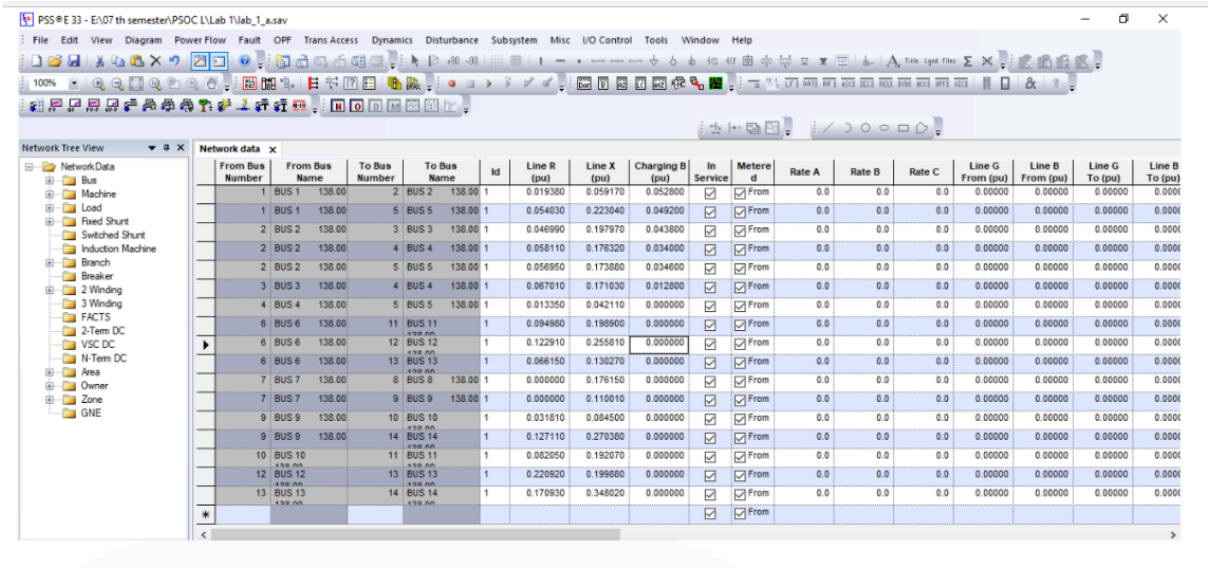
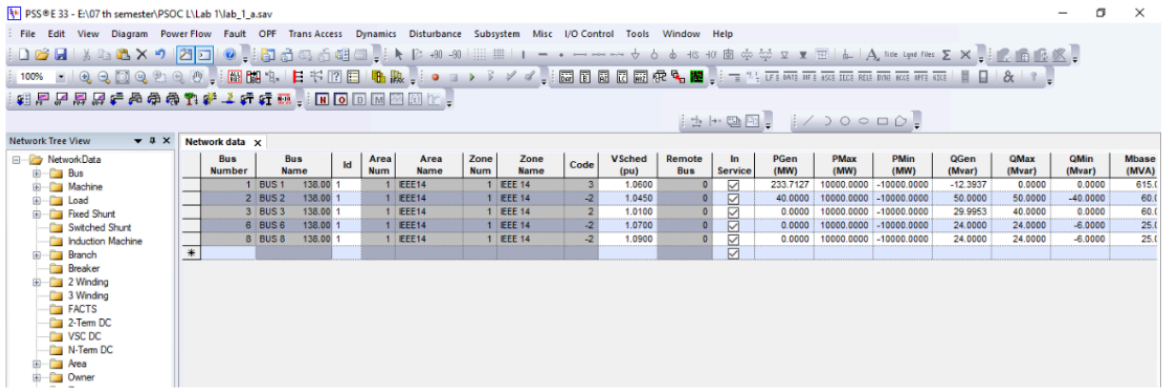
Screenshots of results (bus voltages, line flows, generator outputs).



Bus Number	Bus Name	Base kV	Area Num	Area Name	Zone Num	Zone Name	Owner	Owner Name	Code	Voltage (pu)	Angle (deg)	Normal Vmax (pu)	Normal Vmin (pu)	Emergency Vmax (pu)	Emergency Vmin (pu)
1	BUS 1	138.0	1	IEEE14	1	IEEE 14	1	1	3	1.0600	0.00	1.1000	0.9000	1.1000	0.9000
2	BUS 2	138.0	1	IEEE14	1	IEEE 14	1	1	-2	1.0443	-5.01	1.1000	0.9000	1.1000	0.9000
3	BUS 3	138.0	1	IEEE14	1	IEEE 14	1	1	2	1.0100	-12.83	1.1000	0.9000	1.1000	0.9000
4	BUS 4	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0100	-10.28	1.1000	0.9000	1.1000	0.9000
5	BUS 5	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0125	-8.76	1.1000	0.9000	1.1000	0.9000
6	BUS 6	138.0	1	IEEE14	1	IEEE 14	1	1	-2	1.0515	-14.51	1.1000	0.9000	1.1000	0.9000
7	BUS 7	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0428	-13.40	1.1000	0.9000	1.1000	0.9000
8	BUS 8	138.0	1	IEEE14	1	IEEE 14	1	1	-2	1.0819	-13.40	1.1000	0.9000	1.1000	0.9000
9	BUS 9	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0249	-15.05	1.1000	0.9000	1.1000	0.9000
10	BUS 10	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0219	-15.25	1.1000	0.9000	1.1000	0.9000
11	BUS 11	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0329	-15.01	1.1000	0.9000	1.1000	0.9000
12	BUS 12	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0294	-15.31	1.1000	0.9000	1.1000	0.9000
13	BUS 13	138.0	1	IEEE14	1	IEEE 14	1	1	1	1.0184	-15.15	1.1000	0.9000	1.1000	0.9000
14	BUS 14	138.0	1	IEEE14	1	IEEE 14	1	1	1	0.9590	-15.03	1.1000	0.9000	1.1000	0.9000



From Bus Number	From Bus Name	To Bus Number	To Bus Name	Id	Line R (pu)	Line X (pu)	Charging B (pu)	In Service	Metered	Rate A	Rate B	Rate C	Line G From (pu)	Line B From (pu)	Line G To (pu)	Line B To (pu)
1	BUS 1 138.00	2	BUS 2 138.00	1	0.019380	0.059170	0.052800	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
1	BUS 1 138.00	5	BUS 5 138.00	1	0.054030	0.223040	0.049200	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
2	BUS 2 138.00	3	BUS 3 138.00	1	0.046990	0.197970	0.043800	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
2	BUS 2 138.00	4	BUS 4 138.00	1	0.058110	0.176320	0.034000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
2	BUS 2 138.00	5	BUS 5 138.00	1	0.056950	0.173880	0.034600	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
3	BUS 3 138.00	4	BUS 4 138.00	1	0.067010	0.171030	0.012800	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
4	BUS 4 138.00	5	BUS 5 138.00	1	0.013350	0.042110	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
6	BUS 6 138.00	11	BUS 11 138.00	1	0.094980	0.198900	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
6	BUS 6 138.00	12	BUS 12 138.00	1	0.122910	0.255810	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
6	BUS 6 138.00	13	BUS 13 138.00	1	0.066150	0.130270	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
7	BUS 7 138.00	8	BUS 8 138.00	1	0.000000	0.176150	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
7	BUS 7 138.00	9	BUS 9 138.00	1	0.000000	0.110010	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
9	BUS 9 138.00	10	BUS 10 138.00	1	0.031810	0.084500	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
9	BUS 9 138.00	14	BUS 14 138.00	1	0.127110	0.270380	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
10	BUS 10 138.00	11	BUS 11 138.00	1	0.082050	0.192070	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
12	BUS 12 138.00	13	BUS 13 138.00	1	0.220920	0.199880	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
13	BUS 13 138.00	14	BUS 14 138.00	1	0.170930	0.348020	0.000000	☑	☑ From	0.0	0.0	0.0	0.00000	0.00000	0.00000	0.000
								☑	☑ From							



Bus Number	Bus Name	Id	Area Num	Area Name	Zone Num	Zone Name	Code	VSched (pu)	Remote Bus	In Service	PGen (MW)	PMax (MW)	PMin (MW)	QGen (Mvar)	QMax (Mvar)	QMin (Mvar)	Mbase (MVA)
1	BUS 1 138.00	1	1	IEEE14	1	IEEE 14	3	1.0600	0	☑	233.7127	10000.0000	-10000.0000	-12.3937	0.0000	0.0000	615.
2	BUS 2 138.00	1	1	IEEE14	1	IEEE 14	-2	1.0450	0	☑	40.0000	10000.0000	-10000.0000	50.0000	50.0000	-40.0000	60.
3	BUS 3 138.00	1	1	IEEE14	1	IEEE 14	2	1.0100	0	☑	0.0000	10000.0000	-10000.0000	29.9953	40.0000	0.0000	60.
6	BUS 6 138.00	1	1	IEEE14	1	IEEE 14	-2	1.0700	0	☑	0.0000	10000.0000	-10000.0000	24.0000	24.0000	-6.0000	25.
8	BUS 8 138.00	1	1	IEEE14	1	IEEE 14	-2	1.0900	0	☑	0.0000	10000.0000	-10000.0000	24.0000	24.0000	-6.0000	25.
										☑							

Post lab analysis no more than 350 words?

Convergence and Voltage Profile

The power flow solution successfully converged, providing solved voltage magnitudes and angles for the 14-bus system. The system voltage profile is generally healthy, with no buses violating the lower limit of 0.95 p.u. The lowest observed voltage is at Bus 14, which sits at 0.9590 p.u., just above the critical threshold, while the highest voltage is observed at Bus 8 (1.0819 p.u.).

Slack Bus and System Balance

Bus 1, acting as the slack bus, maintains the system balance by generating 233.71 MW of active power. Since Bus 2 is the only other active generation source (providing 40 MW), the slack bus is supplying the vast majority of the requisite power to meet the total system demand and cover transmission line losses ($P_{gen} = P_{load} + P_{loss}$).

Transmission Loading and Contingencies

While specific line flow data (MVA) is not explicitly visible in the branch tab to confirm heavy loading, the theoretical impact of contingencies can be inferred:

- **Dropping a Line:** Removing a transmission line would increase the system's impedance. This typically causes power to redistribute to parallel lines, potentially causing thermal overloads, and results in greater voltage drops at load buses (likely pushing Bus 14 below the 0.95 p.u. limit).
- **Q-Injection:** Injecting reactive power (Q) helps support voltage levels. If a capacitor bank were switched on at a weak point like Bus 14, it would locally boost the voltage magnitude, improving stability and reducing the burden on remote generators to transport reactive power.